

Original Software Publication

EcgScorer: An open source MATLAB toolbox for ECG signal quality assessment

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ABSTRACT

Cardiovascular diseases claim over 17 million lives annually. Prevention involves adopting healthy habits and regular check-ups, ideally outside hospitals to reduce healthcare costs, leveraging telemedicine tools. However, diagnosing CVDs outside hospitals can be challenging due to noise interference in electrocardiograms (ECGs), necessitating the use of Signal Quality Assessment (SQA) systems. This paper presents a MATLAB toolbox for automated ECG Signal Quality Assessment, featuring a novel method. Furthermore, the toolbox can extract up to 37 Signal Quality Indices (SQIs), commonly used as features in machine learning-based SQA. Therefore, our software has the potential to facilitate the healthcare process, resulting in efficient and cost-effective cardiovascular care.

Metadata

Nr	Code metadata description	Please fill in this column
C1	Current code version	V 1.0
C2	Permanent link to code/repository used for this code version	https://github.com/ecgScorer/ecgScorer
C3	Permanent link to reproducible capsule	none
C4	Legal code license	GNU GPLv3
C5	Code versioning system used	none
C6	Software code languages, tools and services used	MATLAB
C7	Compilation requirements, operating environments and dependencies	MATLAB 2022b and the following toolboxes: signal processing, Statistics and Machine Learning, and Parallel Computing.
C8	If available, link to developer documentation/manual	https://github.com/ecgScorer/ecgScorer/blob/main/README.md
C9	Support email for questions	ecgScorer@gmail.com

1. Motivation and significance

Cardiovascular diseases (CVDs), including coronary heart disease, stroke, and peripheral artery disease, are the world's leading cause of death. In 2019, CVDs were estimated to have claimed 17.9 million lives,

accounting for nearly a third of all deaths worldwide. Typically, CVDs go unnoticed as they lack visible symptoms, and individuals may remain unaware of their condition until a serious event, such as a heart attack or stroke, occurs. Therefore, regular check-ups and screenings are crucial for early detection of CVDs [1,2].

CVDs are commonly detected using an electrocardiogram (ECG), which records the electrical activity of the heart by measuring voltages from electrodes placed at specific positions on the patient's body [3]. Standard ECGs typically last for 10 s and are recorded while the patient is lying down and resting. However, since most cardiac arrhythmias are transient and paroxysmal, long-term ECG monitoring is often necessary to capture arrhythmias during daily activities. Consequently, wearable ECG devices have gained popularity due to their comfort and affordability, offering a unique opportunity to expand telemedicine and improve healthcare access in remote areas [4].

However, ECGs acquired outside of professional settings are prone to noise issues caused by improper electrode placement, muscle movement during prolonged recordings, or environmental factors like electrical interference. These noises can significantly impact the accuracy of automated arrhythmia detection programs, potentially leading to misdiagnosis and incorrect treatment decisions. Therefore, developing robust Signal Quality Assessment (SQA) algorithms is crucial to ensure reliable interpretation of ECG data acquired outside of controlled settings. These algorithms act as gatekeepers, allowing only high-quality

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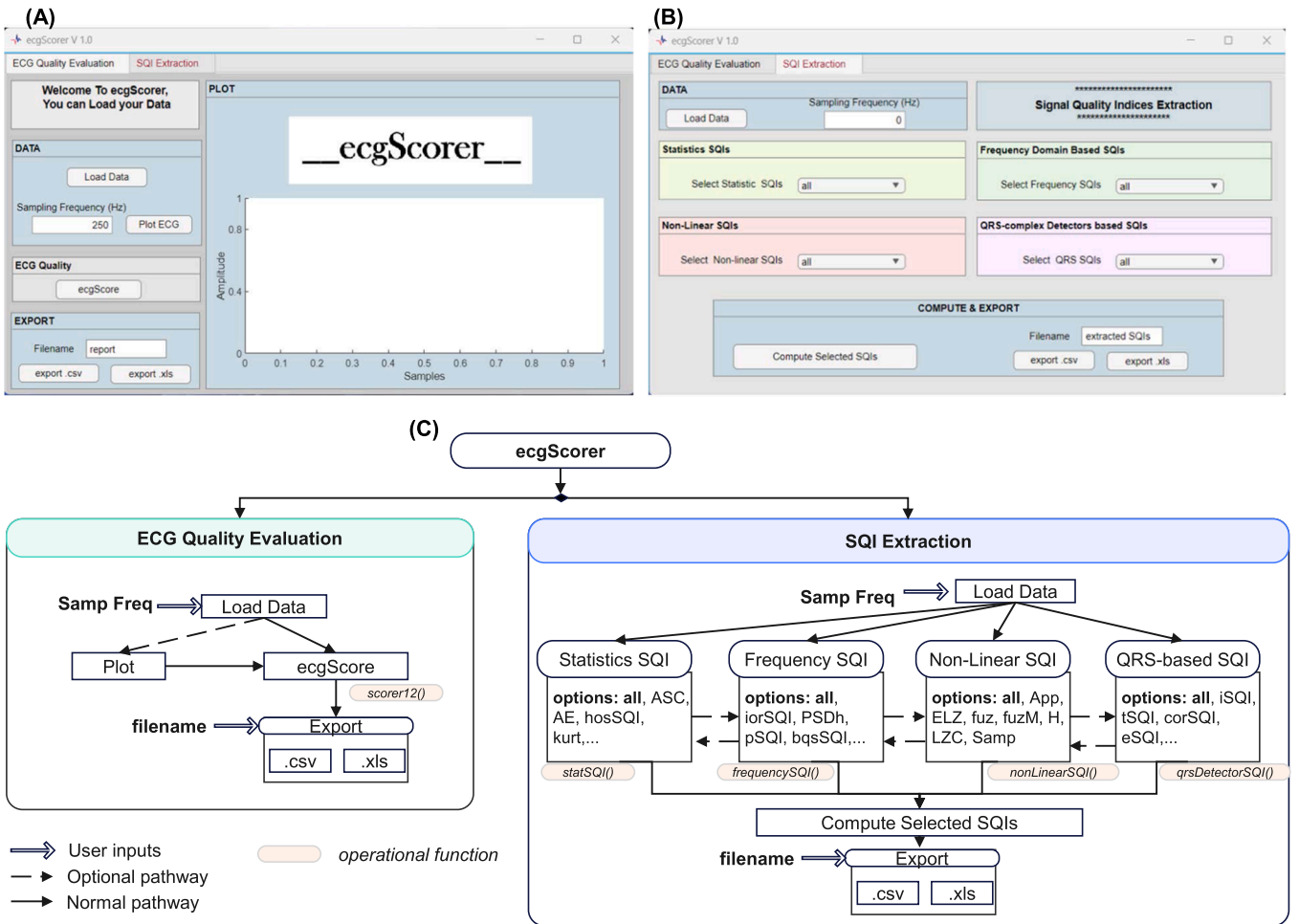


Fig. 1. Structure of the *ecgScorer* app: (A) ECG Quality Evaluation tab, (B) SQI Extraction tab, and (C) App workflow.

segments to be analysed by automated systems [5,6].

SQA systems are used to evaluate the clinical usability of an ECG. For a portion of the signal, instances of artefact (or "noise") are identified and the vital sign measurements for that portion are ignored or corrected/enhanced [5]. Most of the proposed SQA systems in the literature use combinations of rules [7–11], machine learning methods [12–15], and increasingly deep learning methods [16–20]. For most of these methods, the first step is to extract appropriate features that differentiate usable segments from those corrupted by artefacts. These features are called signal quality indices (SQIs) and are used individually with a threshold or combined with a classifier.

Deep learning has emerged as a powerful tool for various tasks in healthcare, including ECG signal quality assessment. Deep learning algorithms can learn complex representations directly from the data, capturing subtle patterns and variations that traditional methods may miss. As a result, deep learning-based methods have shown superior performance in ECG quality assessment tasks [12]. However, a key challenge with existing deep learning models is their limited generalizability. This means they may not perform as well when encountering data from different sources or recording conditions. Additionally, these models are often complex and difficult to deploy on devices used in telemedicine and mobile health (m-health). In these settings, simpler SQA methods based on the fusion of signal quality indices and heuristic rules (SQI-HR) are preferred.

While several researchers have developed simple SQA methods for assessing the quality of ECG signals in telemedicine settings [7,11,11,13, 21–23], the lack of transparency hinders the reproducibility and broader adoption of these methods. Often, the codes are not publicly accessible,

limiting their utility. Only a handful of software packages, such as Neurokit2 [24], PySiology [25], and ECGAssess [26], offer features for ECG quality assessment.

This manuscript introduces a MATLAB toolbox designed specifically for ECG signal quality assessment. This toolbox stands out from previous works in two main aspects. First, it can compute up to 37 different signal quality indices (SQIs). Second, it incorporates a novel signal quality evaluation algorithm that we have previously developed and published [23].

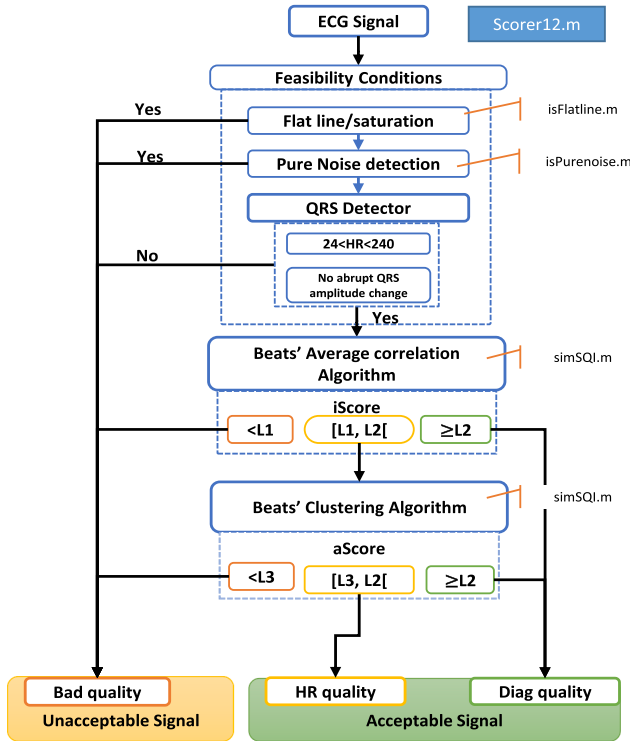
2. Software description

2.1. Software architecture

The *ecgScorer* toolbox is compatible with MATLAB® versions R2022b and later and requires the "Signal Processing" and "Statistics and Machine Learning" toolboxes. The "Parallel Computing" toolbox is also supported but is not mandatory.

The *ecgScorer* toolbox is designed to be modular, flexible, and adaptable to various research workflows. It features a standalone graphical user interface that can be launched with the command '*ecgScorer*'. The interface has two tabs: ECG Quality Assessment and Signal Quality Indices Extraction, as shown in Fig. 1(A) and (B). The Fig. 1(C) present the app workflow.

In the ECG Quality Assessment tab, users can load datasets, compute quality scores, and save the results. The dataset must consist of 10-second ECG recordings with each signal or lead arranged in columns. In the Signal Quality Indices Extraction tab, the user can choose to extract

Fig. 2. Algorithm flowchart of `scorer12` function.

up to 37 Signal Quality Indices (SQIs).

Alternatively, `ecgScorer` can be used in MATLAB's command-line interface or within a script through its functions, allowing for easy interaction with other toolboxes or integration into a larger program.

For illustration purposes, the toolbox includes a dataset of 10 ECG recordings. This data has been extracted from the Computers in Cardiology Challenge 2011 dataset A [27] and comprises five good quality signals and five poor quality signals, which include two with Gaussian noise and three with electrode problems.

2.2. Software functionalities

2.2.1. Signal quality assessment

The `ecgScorer` application enables users to load ECG signal datasets in various formats such as .txt, .csv, .xls, or .dat. The `ecgScorer` calls the `scorer12` function, which requires the sampling frequency of the signals in the dataset. If the frequency is not provided, the default value of 250 Hz is used. The application assesses the quality of the ECG signal and automatically saves the results to the 'base' workspace with the name 'report'. Users can save the report as a CSV or XLS file with a custom name. If no name is provided, the default name will be 'report'. The report matrix comprises five columns: signal number, class, comment, aScore, and iScore. The classes are categorized as '1' for bad signals, '0.5' for intermediate quality signals, and '0' for good quality signals. Fig. 2 illustrates the three main stages of the algorithm used in the `scorer12` function:

(1) The feasibility conditions check:

- **Flat Lines and Saturation Detection:** The function `isFlatline` assesses flat lines or saturated signals and returns 1 if a flat line is detected, and 0 if not. The program normalizes the signal between 0 and 50, considering a sequence of samples as constant when the first derivative equals -1, 0, or 1. Saturation or a flat line is identified when the duration exceeds 1.5 s.
- **Pure Noise Detection:** We employed the function `isPurenoise` to detect pure noise (Gaussian noise (GN)). This

Table 1
Signal Quality Indices in the toolbox.

Feature type	Function names	Option (SQIs)
Statistics-based SQIs	<code>statSQI.m</code>	'ASC' compute the absolute sum of change of the timeseries
		'AE' compute the absolute energy of the timeseries
		'hosSQI' the higher order statistics introduced by Nardeli et al. [21]
		'kurt' the kurtosis
		'LSAM' the longest strike after mean
		'MAC' the mean of absolute change
		'max' the maximum of the timeseries
		'max-min' the difference between the max and the min of the timeseries
		'max-mean' the difference between the max and the mean of the timeseries
		'min' the smallest value of the timeseries
Frequency-based SQIs	<code>frequencySQI.m</code>	'mean' the average value of the timeseries
		'nzc' the zero-crossing rate
		'skn' the skewness
		ST the sum over the timeseries
		'std' the standard deviation of the signals
		'var' the variance of the signals
		'iorsqi' in-out QRS band ratio
		'psdh' the power of the high frequency signals between 10 and 1000 Hz
		'psqi' the relative power in the QRS complex
		'bassqi' the relative power in the baseline
Non-linear based SQIs	<code>nonLinearSQI.m</code>	'hpsqi' the power in 40~100 Hz frequencies to the power of the entire signal
		'Lpsqi' the power spectral density in the low frequency (0~3 Hz) band to the power spectral density in the overall signal
		'mpsqi' the power in 5~35 Hz frequencies to the power of the entire signal
		'psd1' the power spectral density in the low frequency band 0~1 Hz
		'psdnn' Power of ECG signals between 0.05 and 100 Hz
		'app' the approximate entropy
		'elz' the encoding Lempel-Ziv complexity
		'fuz' the fuzzy entropy
		'fuzM' the fuzzy measure entropy
		'h' the Hurst exponent
QRS detectors based SQIs	<code>qrsDetectorSQI.m</code>	'lzc' the Lempel-Ziv complexity
		'samp' the sample entropy
		'isqi' the ratio of the 15th percentile value to the 85th percentile value of the RR intervals
		'tsqi' average correlation coefficient between beats
		'corSQI' average correlation coefficient between the detected beats and the average beats.
		'esqi' The relative energy of the QRS complex
		'pcasqi' the sum of the eigenvalues associated with the five principal components to the sum of all eigenvalues

function removes any baseline oscillation of the signal, rescales it to the range [-1, 1], and computes the number of zero crossings (nzc). If nzc is greater than or equal to 200, the signal is a pure noise.

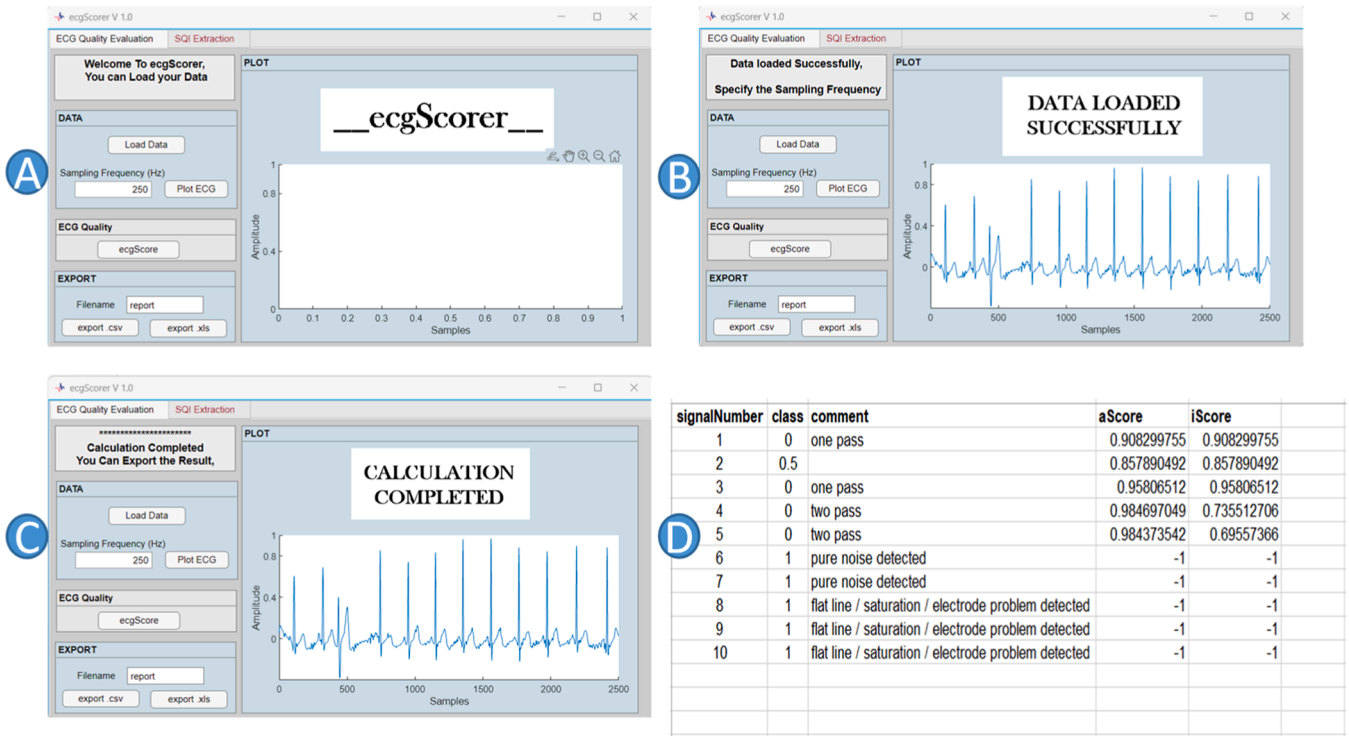


Fig. 3. Screenshots Illustrating ecgScorer App Workflow: (A) the main window, (B) loading and plotting data, (C) calculation of the score quality, (D) report generated after the analysis.

- **Conditions based on detected QRS:** The SQA employs the Pan-Tompkins (PT) algorithm to extract RR intervals and the amplitude differences of detected R-peaks. The initial condition checks the feasibility of the heart rate (HR), aiming for HRs between 24 and 240 bpm, corresponding to 4 and 40 detected beats in a 10-second signal. Additionally, the maximum time difference between two adjacent beats should not exceed 2.5 s.
- (2) **The Beats' average correlation algorithm:** it's the first part of the `simSQI` function. It aims to assess the morphology consistency of any two detected beats within the considered segment. At the end of this step, we have the initial score "iScore" of the segment.
- (3) **The Beats' Clustering algorithm:** it's the second part of the `simSQI` function. It looks for similarities between different QRS complexes and clusters them. This step adjusts the initial score (aScore) for segments containing mixed beats, noises, or both.

2.2.2. Signal quality indices extraction

Table 1 presents the 37 signal quality indices (SQIs) available in the toolbox, categorized into four groups:

Statistical SQIs: 16 indices computed by the function `statSQI.m`.

Frequency Domain SQIs: 9 indices derived using `frequencySQI.m`.

Non-linear SQIs: 7 indices calculated with `nonLinearSQI.m`.

QRS Detector SQIs: 5 indices obtained through `qrsDetectorSQI.m`.

Each SQI can be calculated as an option within its respective category function. Users have the flexibility to calculate one, some, or all of the SQIs. Each function can process single or multiple lead signals, with leads arranged in columns. For detailed information on how to use each function, one can refer to the software documentation.

3. Illustrative examples

Fig. 3 presents a series of screenshots illustrating the 'ECG quality evaluation' functionality of ecgScorer. In Fig. 3(A), we can see the app's

home screen, featuring a 'Load Data' button used to import the ECG signal database the ECG signal database for analysis. After successfully importing the data, the app confirms with a message: 'Data loaded successfully'. The user then enters the sampling frequency and has the option to visualize a sample of the loaded dataset, as shown in Fig. 3(B). Fig. 3(C) displays the interface after pressing the "ecgScorer" button and completing the analysis. Finally, Fig. 3(D) presents the generated report with columns for 'Signal Number,' 'Class,' 'Comment,' 'aScore,' and 'iScore,' detailing the dataset's signal order, quality classification, computational notes, and quality scores.

We provide a video that explains step by step the software functionalities. The video is available at the following link: <https://youtu.be/Jl3DzkBGx6g>.

4. Impact

The ecgScorer toolbox is a valuable resource for researchers working in the field of ECG signal analysis. It can extract 37 signal quality indices (SQIs), which are essential features for many machine learning-based signal quality assessment (SQA) systems. In our previous study, ecgScorer facilitated a comprehensive comparison of various SQI methods found in the literature, enabling a deeper understanding of the strengths and limitations of different approaches [6]. The toolbox provides researchers with the capability to reproduce our experiments and to explore more sophisticated SQA systems. For instance, they can investigate novel deep learning architectures [20] or evaluate the effectiveness of different feature selection techniques for SQA tasks [28].

Furthermore, our software incorporates a novel algorithm specifically designed for automated clinical usability evaluation of ECG signals. The algorithm achieved an average sensitivity of 94.59%, a specificity of 98.38%, and an accuracy of 97.10%, outperforming simple heuristic rule, machine learning, and deep learning methods [23]. This functionality will enable healthcare professionals to focus on usable segments during long-term ECG analysis, thereby reducing their workload. Future work could explore integrating ecgScorer with e-health

systems to enhance continuous and remote monitoring of ECGs, particularly for home care.

5. Conclusions

This paper introduced ecgScorer, which fills a critical gap in ECG analysis by providing a transparent and user-friendly toolbox for quality assessment. This tool is based on a robust signal quality assessment (SQA) algorithm, enabling healthcare professionals to perform rapid evaluations of ECG signals. Additionally, it supports researchers by extracting up to 37 signal quality indices (SQIs), facilitating the development of new SQA methods. The open-source nature of ecgScorer encourages wider adoption and continuous improvement, while its compatibility with MATLAB saves valuable time and increases efficiency. Consequently, the ecgScorer toolbox has the potential to significantly enhance telemedicine.

As a limitation, the current software is designed exclusively for electrocardiogram (ECG) signal analysis. In remote patient monitoring scenarios, additional physiological data, like photoplethysmogram (PPG), are often collected alongside ECG. Further research will focus on extending the capabilities of the software to encompass these additional signal types, thereby broadening its application and utility in remote patient monitoring.

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CRedit authorship contribution statement

Noura Alexandre: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Project administration, Methodology, Investigation. **Fotsing Kuetche:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Ntsama Eloundou Pascal:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Conceptualization. **Simo Thierry:** Writing – review & editing, Visualization, Validation, Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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